Cristobal Chumaceiro Rangel

MSc Computer Science Student, Queen Mary University of London, London, United Kingdom
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RESEARCH INTERESTS

Predictive modelling of environmental systems; scalable machine learning for ecological monitoring, climate risk, and Earth observation.

EDUCATION

Queen Mary University of London — London, United Kingdom
Master of Science in Computer Science — **Distinction** (Projected, 79% average) 2025 – 2026
Bachelor of Science in Business Management — **First Class Honours** 2016 – 2019

PUBLICATIONS

- **C. Chumaceiro Rangel**, S. Poslad, and A. A. Mercado-Velazquez, "Reference-Free Hygroscopic Calibration of Low-Cost IoT Networks for Forecasting Indoor Particulate Matter Exposure," 2026. *[In Preparation].*

RESEARCH PROJECTS

Forecasting Indoor Air Pollution via IoT Sensor Networks (MSc Dissertation) Sep 2026

- Designing an end-to-end data pipeline to calibrate low-cost IoT air quality sensors, correcting for weather-induced hardware distortions using historical meteorological data.
- Developing and benchmarking predictive machine learning models (LSTM neural networks and XGBoost) to forecast localised indoor particulate matter ($PM_{2.5}$) exposure based on outdoor environmental events.

Causal Discovery & Structure Learning in Marine Ecosystems Apr 2026

- Investigating complex ecological interactions by applying Causal Machine Learning to the 1980–2020 Global Coral-Bleaching Database (GCBD).
- Comparing multiple structure learning algorithms to model environmental dependencies and generating Directed Acyclic Graphs (DAGs) to visualise causal pathways in coral bleaching events using Bayesys software.

Predicting Surface Shortwave Radiation Mar 2026

- Evaluated Linear Regression and K-Nearest Neighbours (KNN) algorithms for forecasting surface shortwave radiation using an 11-year meteorological dataset.
- Optimised models through feature selection, hyperparameter tuning, and cross-validation to ensure real-world generalisability.

SOFTWARE PROJECTS

User-Centred Design of a Collaborative Research Interface Apr 2026

- Led a 9-person team in designing the user interface for a collaborative research application, culminating in a high-fidelity prototype, an academic report, and a formal presentation.
- Directed the end-to-end HCI lifecycle through user research, persona creation, and an iterative development process driven by qualitative user and expert evaluation.

Real-Time Transit Database Architecture Dec 2025

- Designed and implemented an Oracle database application to manage real-time passenger journey data, engineering complex SQL queries for anomaly detection and fare tracking.
- Developed comprehensive test data to verify query functionality across edge cases.

Personal Fitness Tracker Jan 2025

- Built a full-stack fitness tracking web application using Python, JavaScript, and SQLite.

SKILLS

- **Languages:** Python, R, SQL, C, JavaScript, HTML, CSS
- **Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch
- **Soft Skills:** Effective Communication, Initiative, Critical Thinking, Adaptability, Interdisciplinary Collaboration

HACKATHONS & COMPETITIONS

- **Francis Crick Institute Innovation Challenge:** Jul 2026
  Pitched a computer vision platform mapping meal data to continuous glucose monitors.
- **Encode Club Hackathon** *(Finalist & BGA Bounty Winner)*: Jun 2026
  Developed a federated learning platform for verifiable carbon credits.
- **Rockstar Games Datathon:** Nov 2025
  Conducted exploratory data analysis in Python to model gameplay behaviour and anomalies.
- **Nationwide Building Society Hackathon** *(1st Place)*: Oct 2025
  Designed system architecture and data pipelines for personalised air quality monitoring.

PROFESSIONAL EXPERIENCE

Dover Street Market London, United Kingdom
Client Services Operations Manager (Previously Floor Manager / Sales) 2019 – Present

- Progressed through four roles to manage a 5-person operations team, coordinating cross-departmental logistics and leading daily strategic stand-ups.

OTHER EXPERIENCE

Queen Mary University of London London, United Kingdom
Language Speaking Practice Volunteer 2025 – 2026

- Ran conversational Spanish sessions to foster cross-cultural communication among QMUL students.

Pokémon Society President 2018 – 2019

- Led operations, budgeting, and marketing for a 60-member society, successfully securing corporate sponsorships.

REFERENCES

Available upon request.